

DEEP REINFORCEMENT LEARNING

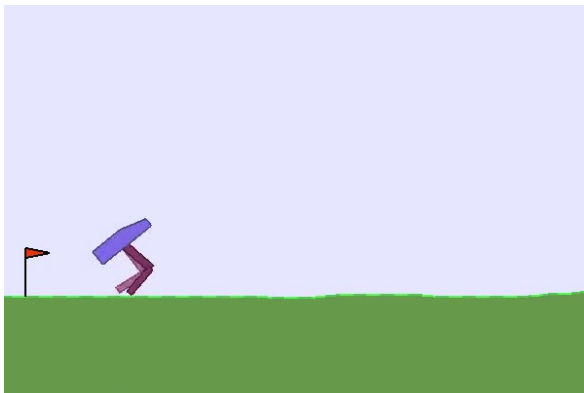
365.382 (SS26)

Assignment II: Learning to Walk with PPO



LIT AI LAB
Institute for Machine Learning

Learning to Walk Again



[BipedalWalker Video](#), [Gym Webpage](#)

Recommendations

- You should see some progress in about 15 min
- Handle high variance in your reward (range < -100 and $> +300$)
- Optionally impl. GAE ¹
 - ↔ computes balanced Advantage
- Try shallow network architectures
- Play around with hyperparameters after you have something working
- Change only one thing at a time and try out an experiment
- The critic has to learn faster than the actor

¹Schulman, J. et al. (2016) High-dimensional continuous control using generalized advantage estimation. *In Proceedings of the International Conference on Learning Representations (ICLR)*.

Hints - PPO

- Again: **Transformers are discouraged here!**
 - ↪ Deep models won't help.
- Agent is not getting better?
 - ↪ Talk with your colleagues and post on forum. Please start soon.
- To get you started:
- Check-out evaluation script for hints on action selection.

Submission

- Submission server: <https://apps.ml.jku.at/challenge>
- Export your trained agent as ONNX file
- Upload ONNX file to submission server
 - ☐ Code submission closes on **Friday, May 1st, 18:00**.
 - ☐ Max model file size: **50MB**.
- Submit your **code**, **best model** and a short **report** to Moodle
 - ☐ Report the performance on your local setting.
 - ☐ Moodle submission stays open 24h after the challenge closes.
- Please upload a zip file named `k<studentid>.zip` containing:
 - ☐ `code.ipynb` / `code.py`
 - ☐ `model.onnx`
 - ☐ `report.pdf`

Evaluation

- < -50 : 0 Pts.
- ≥ -50 : 2 Pts.
- ≥ 50 : 4 Pts.
- ≥ 150 : 6 Pts.
- ≥ 225 : 8 Pts.
- ≥ 275 : 10 Pts.

Presentations

- If you are willing to give a short presentation of your work, fill out the poll Model that closes Sunday evening after the submission deadline.
- Students selected to present will receive +2 bonus points.